

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Cardiogenic Shock Management: A Framework for Individualized Microaxial Flow Pump Support

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ABSTRACT: Cardiogenic shock (CS) is a highly heterogeneous syndrome with persistently high mortality despite advances in therapies, including temporary mechanical circulatory support. The disparate results of trials on temporary mechanical circulatory support in CS suggest that improved strategies, including better patient selection, are needed in the context of CS heterogeneity. Notably, a microaxial flow pump device has recently shown a survival benefit in ST-segment–elevation myocardial infarction-associated CS in the DanGer Shock trial (Danish-German Cardiogenic Shock), marking a positive outcome amid many negative or neutral trials. As CS phenotyping becomes increasingly important for developing tailored therapeutic strategies, it is essential to identify factors contributing to variable phenotypes, including shock severity, cause, hemodynamic profile, ventricular function, shock pattern, and patient-centered factors that may influence microaxial flow pump candidacy. Successful CS management with a microaxial flow pump may be best guided by an individualized and proactive approach, tailored to each patient's risk profile and clinical course. It is rational to implement this strategy at centers capable of providing and managing advanced temporary mechanical circulatory support options while determining candidacy for advanced heart failure therapies. Our review underscores the importance of CS phenotyping and tailoring temporary mechanical circulatory support—particularly microaxial flow pump support—to patient-specific factors that are central to optimizing outcomes in CS management.

Key Words: disease progression ■ heart failure ■ patient selection ■ phenotype ■ shock, cardiogenic ■ ventricular function

Cardiogenic shock (CS) is a heterogeneous, highly morbid, and often fatal syndrome characterized by end-organ hypoperfusion due to reduced cardiac output (CO). Despite advances in monitoring and therapeutic tools, mortality remains unacceptably high (27%–51%).^{1,2} Beyond the high risk of in-hospital mortality, the incidence of long-term morbidity among CS survivors is a major concern: 42% require greater medical support beyond baseline, 47.5% are readmitted, and 15.3% die within 1 year of discharge.³ The rising clinical burden, evidenced by a 3.34-fold increase in CS hospitalizations from 2004 to 2018,² emphasizes the importance of countering with clear frameworks for CS phenotyping, risk stratification, and tailored treatment strategies.

Recent advances in temporary mechanical circulatory support (tMCS) have reshaped the management of CS. Among available platforms, Impella (Abiomed, Danvers, MA) is a transvalvular microaxial flow pump (mAFFP) that unloads the left ventricular (LV) and augments systemic and coronary perfusion. Since its initial US Food and Drug Administration approval in 2008 for high-risk percutaneous coronary interventions (PCI), Impella use has steadily expanded to include CS. Among acute myocardial infarction–related CS (AMI-CS), Impella utilization rose from 9.1% in 2015 to 21.5% in 2021.⁴ More recently, in 2024, the randomized DanGer Shock trial (Danish-German Cardiogenic Shock) demonstrated a significant reduction in 6-month mortality with routine Impella CP use on top of standard care versus standard care alone in selected

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Nonstandard Abbreviations and Acronyms

AMI	acute myocardial infarction
CO	cardiac output
CS	cardiogenic shock
DanGer Shock	Danish-German Cardiogenic Shock
HF	heart failure
LV	left ventricular
mAFP	microaxial flow pump
ML	machine learning
PAC	pulmonary artery catheter
PCI	percutaneous coronary intervention
RV	right ventricular
SCAI	Society for Coronary Angiography and Interventions
STEMI	ST-segment–elevation myocardial infarction
tMCS	temporary mechanical circulatory support

ST-segment–elevation AMI-CS patients.⁵ Reflecting this emerging evidence, the 2025 American College of Cardiology/American Heart Association/American College of Emergency Physicians/National Society of EMS Physicians/Society for Cardiovascular Angiography and Interventions (SCAI) guidelines for the management of patients with acute coronary syndrome recommend mAFP support for selected ST-segment–elevation myocardial infarction (STEMI) AMI-CS patients.⁶ These recent developments signal a growing role for mAFP in contemporary CS care. Nevertheless, given the high complication rates associated with mAFP and other large-bore tMCS devices, their use should be judiciously targeted to patients most likely to benefit.^{7–9}

Despite ongoing innovation and a broader therapeutic landscape, improving CS survival remains a pressing, unmet need that calls for more effective, patient-specific strategies. While there is insufficient clinical evidence to demonstrate that the treatment effect of mAFP varies by patient characteristics or that tailored therapeutic strategies improve outcomes, the substantial phenotypic variation in CS that shapes presentation and trajectory may limit the effectiveness of a one-size-fits-all approach.^{10–15} Beyond the underlying cause, a host of intrinsic and extrinsic factors, including shock severity (SCAI stage), hemodynamic profile (congestion status, ventricular involvement LV, right ventricular [RV], and biventricular), shock phenotype (classic vasoconstricted or mixed), and patient-centered considerations (eg, frailty), shape presentation and may necessitate a tailored management strategy.

Two questions are paramount: (1) How can we better select patients with heterogeneous presentations of CS for tMCS? and (2) How can emerging therapies, particularly mAFP, be tailored to individual patients while mitigating associated complications? This review outlines a phenotype-driven framework to personalize and optimize mAFP therapy across evolving CS phenotypes and offers practical guidance for clinical implementation.

MICROAXIAL FLOW PUMPS IN CS MANAGEMENT

Left-sided mAFPs, such as Impella devices (CP and 5.5; Abiomed, Danvers, MA), are percutaneous, catheter-mounted mAFPs positioned across the aortic valve, with the inlet in the LV and the outlet in the ascending aorta. This modality is distinct among current commercially available tMCS options in that it provides active, continuous antegrade flow across the valve with direct LV unloading. By delivering antegrade flow from the LV to the aorta, these devices augment systemic CO, improve mean arterial pressure, reduce LV end-diastolic pressure, and enhance coronary perfusion pressure.^{16–19}

Among the available left-sided mAFPs, each provides a different degree of hemodynamic support and requires a distinct insertion strategy. For example, the smaller Impella CP is typically placed percutaneously via the femoral artery, delivering up to ≈3.7 L/min of flow. In contrast, the larger Impella 5.5 is capable of delivering up to 5.5 L/min of flow and is usually inserted via the axillary artery through a surgical cutdown. Newer mAFPs are currently under investigation but have yet to be approved for commercial use. A comparative summary of the intended hemodynamic benefits, limitations, and contraindications of mAFP (eg, Impella) and other left-sided tMCS platforms is provided in Table 1.

Alternatively, right-sided mAFPs provide right ventricle (RV) support for patients with CS and are inserted via the right internal jugular/subclavian vein (eg, Impella RP-Flex) or the femoral vein (eg, Impella RP). They unload the RV by maintaining continuous antegrade flow from the junction of the right atrium and superior or inferior vena cava to the main pulmonary artery. Much less is known about the clinical outcomes associated with right-sided mAFPs, but these devices can be used for single-ventricle RV support or incorporated into a biventricular support strategy with left-sided mAFPs (Bi-Pella).

PHENOTYPE-DRIVEN CS MANAGEMENT USING MICROAXIAL FLOW PUMPS

In general, failure to achieve hemodynamic and perfusion targets in CS, or a high likelihood that they will not be met with medical therapy alone, should prompt consideration of tMCS initiation. With growing recognition of CS phenotyping

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Table 1. Anticipated Hemodynamic Benefits, Limitations, and Contraindications of mAFP (eg, Impella) and Other Forms of LV tMCS

tMCS device	Mechanism of support	Typical level of support	Fr Caliber	Advantages	Limitations	Contraindications
IABP	Aortic counterpulsation: LV unloading by afterload reduction (balloon deflation); improved CPP by increase MAP and diastolic blood pressure (balloon inflation)	Generally 0.5–1 L/min of support but may be variable	7–8 Fr arterial sheath	Smaller bore catheter required for placement Directly ↑ CPP while unloading the LV Decreases LV oxygen consumption Afterload reduction Venting strategy option for VA-ECMO May be used without systemic anticoagulation if support ratio is 1:1	Limited degree of hemodynamic support in comparison to other tMCS (0.5–1 L/min) Isolated LV support	Aortic regurgitation Aortic dissection Uncorrected bleeding diathesis Severe PAD
Impella (Abiomed, Danvers, MA)	Microaxial continuous flow pump from LV to Aorta that offers direct LV unloading and improved systemic perfusion	Impella CP—Up to 3.7 L/min of support Impella 5.5—Up to 5.5 L/min of support	14 Fr arterial sheath (Impella CP) 23 Fr arterial sheath (Impella 5.5)	Targeted multifaceted hemodynamic benefits: ↑↑ CO, MAP, and CPP while ↓↓ LVEDP Opportunities for use in protected PCI or EP ablation procedures Options for nonfemoral left-sided support (Impella 5.5) Venting strategy option for VA-ECMO patients	Elevated complication risk (bleeding, hemolysis) less support offered by Impella CP Femoral placement routinely used for Impella CP, limits mobility Impella CP and 5.5 only support LV	LV thrombus Mechanical aortic valve Severe PAD Inability to tolerate anticoagulation Severe RV failure
VA-ECMO	Continuous centrifugal flow pump from venous system to aorta (various configurations)	Up to 2–7 L/min of support	15–24 Fr arterial sheath 19–25 Fr venous sheath	Biventricular support Capacity to modulate gas exchange Large range of titratable support Wide variety of VA-ECMO configurations	↑↑ afterload state that may require additional tMCS for LV unloading (eg, ECPella) Risk of cerebral hypoxia High bleeding and stroke risk	Small body size Severe PAD Aortic dissection Moderate to severe aortic valve insufficiency Inability to tolerate anticoagulation No exit strategy
VA-ECMO+LV venting with IABP or Impella (ECPella)	LV venting using Impella or IABP counterpulsation aims to unload the LV in the high afterload state created by retrograde aortic flow in VA-ECMO	Up to 2–7 L/min of support	As above	Biventricular support Capacity to modulate gas exchange Large range of titratable support Enables increases myocardial O ₂ supply and decreases myocardial O ₂ demand LV venting strategy	LV unloading with Impella 2.5/CP was not associated with increased mortality when compared with LV unloading with IABP but was associated with increased adverse events ²⁰	See Impella and VA-ECMO contraindications above

CO indicates cardiac output; CPP, coronary perfusion pressure; ECPella, VA-ECMO plus Impella; IABP, intra-aortic balloon pump; LA, left atrium; LV, left ventricular; LVEDP, left ventricular end-diastolic pressure; mAFP, microaxial flow pump; MAP, mean arterial pressure; tMCS, temporary mechanical circulatory support; PAD, peripheral arterial disease; PCI, percutaneous coronary intervention; RV, right ventricle and VA-ECMO, veno-arterial extracorporeal membrane oxygenation.

and expanding tMCS use, optimal tMCS deployment should integrate shock severity, cause, ventricular involvement, hemodynamic patterns, expected clinical trajectory, patient-centered factors, and institutional expertise. Figure 1 presents an example of a phenotype-driven framework for mAFP candidacy and device selection. While an array of variables influences initiation decisions, the overarching question remains whether support will serve as a bridge to myocardial recovery or to durable advanced therapy (ie, durable LV assist device or orthotopic heart transplantation).

Data Gathering to Guide Decision-Making

Timely data capture and a comprehensive synthesis of physical examination findings, laboratory values, imaging findings (particularly echocardiography), and invasive hemodynamic parameters may be useful in informing

decisions on mAFP initiation, escalation, and deescalation, as outlined in Table 2.

A central component of this process is the use of invasive hemodynamic assessment with a pulmonary artery catheter (PAC). In observational studies, early use of a PAC in patients with CS has been associated with improved survival^{22,23} and may be particularly helpful in determining the appropriateness of mAFP. After baseline data are obtained, efforts should focus on normalizing abnormal hemodynamic, laboratory, and clinical parameters with medical therapy (eg, inotropes and vasopressors) and tMCS.

Phenotype-Guided Strategies to Determine Impella Initiation Candidacy

At present, the only evidence-based phenotypic criteria demonstrating a survival benefit with mAFP pertain to

the use of Impella CP in patients with STEMI-related CS who present without RV failure or out-of-hospital cardiac arrest.⁵ However, a broad spectrum of CS phenotypes will continue to be encountered in clinical practice, underscoring the need to account for phenotypic factors, even in the absence of robust evidence. Until such data are available, mAFP use should be guided by a rational, physiology-based approach that accounts for shock severity, cause, hemodynamic profile, and holistic patient factors.

Shock Severity

When evaluating the candidacy of a patient with CS for mAFP therapy, the first essential step is to classify the severity of shock. Frameworks such as SCAI staging are well validated for rapid risk stratification.^{24–28} Beyond prognostication, the SCAI framework provides a dynamic classification that integrates evolving clinical data to guide escalation and therapeutic decision-making.²⁹ Nevertheless, the SCAI framework may be susceptible to subjective interpretation, inspiring ongoing efforts by other groups to standardize and define the stages to enhance universal communication.^{30,31}

For example, the Cardiogenic Shock Working Group has proposed a novel approach to defining SCAI stages by integrating hemodynamic (systolic blood pressure, mean arterial pressure), laboratory (lactate, ALT, pH), and treatment intensity (vasoactive agents, inotropes, tMCS) parameters. This framework aims to provide a rapidly deployable, criterion-driven tool that emphasizes clinical variables for physicians and promotes a common language for severity grading.³¹ Other authors have advocated leveraging lactate clearance to inform decisions surrounding escalation strategies.³² Ultimately, a framework that defines severity using objective parameters may facilitate the development of algorithmic tools to predict prognosis and guide tailored therapeutics. Regardless of the specific criteria used to define the SCAI Shock Classification, it is generally reasonable to consider tMCS initiation or escalation for eligible patients with deteriorating or refractory CS despite medical therapy.

It is also important to acknowledge that CS severity often evolves early after presentation. Data from the Cardiogenic Shock Working Group registry show that patients presenting in SCAI stages B or C often

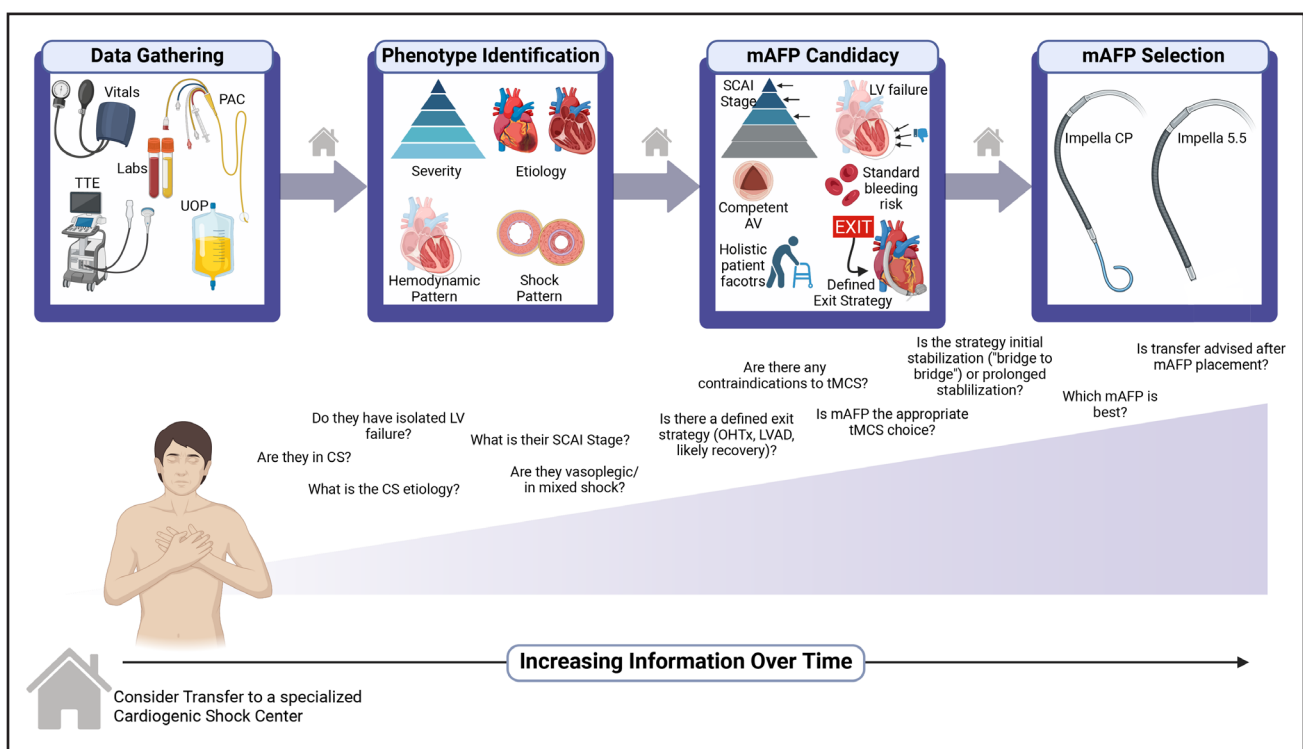


Figure 1. Proposed phases in phenotype-driven management of cardiogenic shock (CS) with microaxial flow pump (mAFP) support.

Data gathering involves a progressive, iterative accumulation of information, including vital signs, laboratory values, invasive hemodynamics (eg, pulmonary artery catheter [PAC] use), imaging (eg, transthoracic echocardiography), and other clinical parameters (eg, urine output). Phenotype identification delineates CS severity, cause, hemodynamic pattern, and shock pattern. The decision to initiate mAFP therapy involves careful assessment of candidacy and contraindications. Finally, mAFP selection should align with the intended purpose (eg, acute stabilization with e.g. Impella CP vs prolonged support with e.g. Impella 5.5, while considering center resources and, if appropriate, the option to transfer to a designated shock center exists at any stage. AV indicates aortic valve; LV, left ventricular; LVAD, left ventricular assist device; OHTx, orthotopic heart transplant; TTE, transthoracic echocardiography; tMCS, temporary mechanical circulatory support; and UOP, urine output. Created in BioRender. May, A. (2026) <https://BioRender.com/lqgchis>.

Table 2. Clinical, Hemodynamic and Laboratory Goals to Inform Decision to Initiate, Escalate or Deescalate mAFP

Clinical parameters	Target values	Notes
Urine output	>0.5 cm³/kg per h*	Markers of perfusion and congestion
Capillary refill	<2 s	
Skin exam	warm, dry, and normal coloration for skin tone	
Mental status	A&Ox4 or trending toward baseline	
Lungs	Clear to auscultation, baseline oxygen requirement	
Lower extremity edema	No edema, improving edema, skin wrinkling	
Juglar venous distension	Improving, may be confounded by tricuspid valvulopathy	
Hemodynamic parameters		
MAP	≥65 mm Hg	Essential for adequate end organ perfusion pressure
SBP	>90 mm Hg	
RAP	<8 mm Hg	Reflects adequate ventricular performance and unloading
PCWP	<15 mm Hg	
CI	>2.2 L/min per m²	Reflects adequate CO
CPO	1.0–1.5 W	Less than 0.6 W is the strongest hemodynamic correlate of mortality in CS
Laboratory parameters		
Serum lactate	<2 mmol/L	Markers of perfusion and congestion
pH	>7.3	
Renal markers: Serum Cr, Cystatin C	Trending toward baseline	
Liver markers: AST/ALT, Bilirubin, INR	Trending toward normal	
Cardiac Markers: Troponin, NT-proBNP	Down trending or static	Markers of myocardial injury and myocardial stretch
ScVO ₂	60–70%	Surrogate marker for CO
Imaging		
Chest X-ray	Improving pulmonary vascular congestion	Can track pulmonary edema severity, assess other contributors to hypoxia, and assist in informing general Impella position
Lung ultrasound	<3 B-lines in a view, ²¹ resolving pleural effusions	Can track pulmonary edema severity May help differentiate etiologies of hypoxia (pleural effusion, pulmonary edema, pneumonia) Rapidly deployable with bedside ultrasound
Transthoracic echocardiogram	Visual assessment biventricular systolic recovery Echocardiographic evidence of normalized filling pressures (reduced E velocity, collapsible IVC) Recovery of objective measurements of systolic function LV: estimated or calculated ejection fraction (%), LVOT-VTI, RV: TAPSE Ideal Impella placement as measured from aortic annulus to Impella inlet Impella CP: 3.5 cm Impella 5.5: 5 cm	Rapidly deployable at patient bedside Enables serial and comprehensive cardiac assessment Enables serial assessment for myocardial recovery and device-related complications (eg, acute valvopathy) Enables serial assessment of Impella positioning (preferred window left parasternal long axis)

A&Ox4 = alert and oriented to person, place, time, and situation; ALT = alanine aminotransferase; AST = aspartate aminotransferase; CI, cardiac index; CO, cardiac output; CP = Cardiac Power; CPO, cardiac power output; CPP, coronary perfusion pressure; CS, cardiogenic shock; INR = International Normalized Ratio; IVC, inferior vena cava; LV, left ventricular; LVOT-VTI, left ventricular outflow track velocity time integral; mAFP, microaxial flow pump; MAP, mean arterial pressure; NT-proBNP = N-terminal pro-B natriuretic peptide; PCWP, pulmonary capillary wedge pressure; RAP, right atrial pressure; RV = right ventricle; SBP, systolic blood pressure; ScVO₂, Central venous oxygen saturation; TAPSE, tricuspid annular plane systolic excursion; and UOP, urinary output.

*UOP should be >0.5 cm³/kg per h as the minimum target in patients who are not receiving diuresis.

deteriorate to more advanced stages within the first 24 hours, a trajectory associated with greater treatment intensity (vasoactive medications and tMCS use) and higher mortality.^{31,33} Their report highlights that the first 24 hours following CS presentation, the golden day of shock, is a dynamic and essential treatment window for defining prognosis through serial assessment of CS severity.³³

Cause

In parallel, incorporating cause as a core element of the patient's phenotype provides critical insight into trajectory. For example, registry data indicate that, despite having fewer comorbidities, patients with more acute etiologies, such as AMI-CS or other causes of de novo acute heart failure (HF; eg, viral myocarditis), often present with more severe shock and have worse in-hospital

outcomes than those with HF-CS.¹⁰ One hypothesis is that patients with de novo presentations may be driven by more inflammation and lack the compensatory mechanisms seen in acute-on-chronic HF, in which greater hemodynamic resilience may confer improved responsiveness to noninvasive therapies (eg, vasoactive agents, inotropes, afterload reduction).^{10,14} Given the high risk of rapid deterioration in de novo acute HF and AMI-CS, timely deployment of tMCS—hopefully as a bridge to recovery—may be especially critical.³⁴

Hemodynamics and Shock Pattern

The selection among tMCS device options should be guided by the overall hemodynamic profile including ventricular involvement as well as perfusion and congestion status. LV congestion, often reflected by a pulmonary capillary wedge pressure >15 mm Hg (a surrogate for elevated LV end-diastolic pressure), is common in CS. Patients with predominant LV failure are well suited to left-sided mAFP (eg, Impella CP, 5.5), which unload the LV to relieve congestion, augment forward flow to support systemic perfusion, and improve coronary perfusion. The ideal candidate for a left-sided mAFP has intrinsic LV dysfunction with systemic hypoperfusion (SCAI stage C or worse) and an RV capable of maintaining adequate LV preload.³⁵ Moreover, in our experience, mAFPs are particularly useful in markedly volume-overloaded CS, as they can raise mean arterial pressure and augment forward flow and renal perfusion pressure. These interventions may enhance diuretic responsiveness, ultimately facilitating decongestion and lowering central venous pressure.^{36,37}

In patients with CS and significant RV failure as evidenced by elevated right atrial pressure, a right atrial pressure-to-pulmonary capillary wedge pressure ratio >0.86,³⁸ a pulmonary artery pulsatility index <1.0,³⁹ or other hemodynamic or echocardiographic signs of RV-to-PA uncoupling,⁴⁰ isolated left-sided mAFP may be less likely to rescue the patient as a standalone therapy. A failing RV that cannot sustain adequate LV preload inherently limits the effectiveness of left-sided support, potentially resulting in suction events after left-sided mAFP placement. Moreover, coexisting hypovolemia can further compromise LV filling and precipitate suction events, provoking frequent device alarms and potential ventricular dysrhythmias from catheter-induced myocardial irritation. In such cases, optimization of medical therapy (eg, judicious fluids, vasopressors, and inotropes) and consideration of right-sided tMCS, such as Impella RP (Abiomed, Danvers, MA), as part of a Bi-Pella configuration (LV Impella+RP Impella), the ProtekDuo dual-lumen cannula (LivaNova, London, United Kingdom), or use of an external pump such as CentriMag (Abbott, Abbott Park, IL), may be necessary. For patients with severe biventricular failure, escalation to venoarterial extracorporeal membrane oxygenation may be

most appropriate, with observational studies suggesting benefits of LV venting with Impella (VA-ECMO plus Impella [ECPELLA]).^{41,42}

As CS phenotyping advances, vasodilatory-CS and mixed shock have emerged as distinct and difficult-to-manage phenotypes.^{12,13,43,44} These phenotypes, characterized by low cardiac output with low systemic vascular resistance, complicate tMCS selection because flow augmentation alone may not directly address the distributive component or its underlying pathophysiology.⁴⁵ In a recent review, Jentzer et al⁴³ classify mixed shock into 3 etiologic categories: cardiogenic-vasodilatory shock, vasodilatory-CS, and primary mixed shock based on the primary shock pathology; each category carries distinct etiologies, potential therapeutic implications, and prognoses. Individuals with mixed shock require interventions to increase cardiac output (which could include tMCS) as well as to restore vascular tone (ie, vasopressors), and tMCS alone is not expected to be adequate. In vasodilatory-CS, higher flows through tMCS may be required to restore perfusion and optimize oxygen delivery. Although data in mixed-shock patients supported with mAFP are limited, a retrospective review of patients with mixed cardiogenic-septic shock supported with VA-extracorporeal life support found that higher flows trended toward improved survival.⁴⁶ Optimal management of this highly heterogeneous and increasingly recognized phenotype remains ill-defined, and further research is needed to refine phenotype classification and clarify the roles of pharmacological therapy and tMCS strategies.

Holistic Patient Evaluation

A holistic patient evaluation is essential before initiating tMCS in CS. Patient-centered factors, including advanced age, frailty, baseline functional status, indeterminate neurological status after cardiac arrest, concurrent acute illness (eg, infection, bleeding), and the cumulative burden of chronic comorbidities, should be weighed carefully, as each materially influences prognosis and is not likely to confer risk that is modifiable by tMCS.⁴⁷ Therefore, tMCS should be reserved for patients with a reasonable likelihood of benefit. The decision to deploy mAFP should follow a rigorous risk-benefit analysis that balances potential complications, such as major bleeding, hemolysis, limb ischemia, infection, and stroke against anticipated hemodynamic and clinical gains. Equally important is establishing a clear therapeutic intent and a well-defined exit strategy before initiation. Clinicians should determine whether the mAFP is intended as a bridge to myocardial recovery or to durable therapy. If myocardial recovery is unlikely, early evaluation for advanced HF therapies is essential, with careful alignment of the patient's goals of care with any potential aggressive interventions (eg, LV assist device implantation or orthotopic heart transplantation).

Microaxial Flow Pump Selection and Placement

The anticipated intensity and duration of hemodynamic support may inform the decision to use 1 mAFP over another. For example, the Impella CP is the most commonly used device in the catheterization laboratory and is primarily used for rapidly establishing hemodynamic support in high-acuity CS cases or during planned high-risk PCI. Its key advantage is the ability to be placed percutaneously, allowing for rapid deployment and urgent stabilization, often serving as a bridge to a bridge in patients presenting with rapidly decompensating CS.

In contrast, the Impella 5.5 is typically reserved for patients with CS who require higher levels of hemodynamic support or a longer duration of support, such as those awaiting durable advanced HF therapies. Beyond providing greater hemodynamic support, the surgical axillary insertion of the Impella 5.5 offers several important advantages. These include lower device-related complication rates, facilitation of earlier mobilization, safer extubation practices,^{48,49} and a lower risk of limb ischemia in patients with severe peripheral artery disease.⁵⁰ However, the smaller size of the axillary artery may limit this approach in smaller patients who cannot accommodate the larger sheaths required by these devices.

Integrating Temporary Mechanical Circulatory Support Into Regionalized Systems of Care

tMCS placement and management are well suited to a hierarchical, coordinated healthcare network that aligns with proposed schemas for stratifying centers by expertise (Figure 2).^{51–54} We consider transthoracic echocardiography and right-heart catheterization essential for the characterization of CS; however, these diagnostics may not always be feasible at some referring hospitals. In such cases, recognition, initial stabilization, and transfer to a destination hospital with greater access to advanced diagnostic and treatment capabilities are warranted.

Patients with CS initially presenting to a referral hospital are typically stabilized in the catheterization laboratory with a transfemoral intraaortic balloon pump or Impella CP. This approach is particularly valuable where access to tMCS or CS management expertise is limited (levels 2 and 3 centers). For example, early use of an intraaortic balloon pump or Impella CP at the referral hospital may facilitate safe transfer to a higher-level destination hospital (level 1 center) for advanced care (eg, high-risk revascularization, structural interventions, or evaluation for durable LV assist device implantation or orthotopic heart transplantation).

On arrival at the destination hospital, patients requiring more robust or prolonged support may be upgraded to a high-support axillary mAFP (eg, Impella 5.5), enabling higher hemodynamic support, early mobilization, and

enough time for comprehensive multidisciplinary evaluation and planning while serving as a bridge to recovery or definitive therapy. At capable centers, standardized assessment, phenotype-guided device selection, and predefined exit strategies can help achieve therapeutic goals, maximize the benefits of tMCS, minimize complications, and optimize long-term outcomes.⁵⁵ This process is best coordinated by a dedicated multidisciplinary shock team that identifies and addresses barriers while systematically evaluating patient candidacy for advanced HF therapies.⁵⁶

MICROAXIAL FLOW PUMP MAINTENANCE, ESCALATION, AND DEESCALATION

Once an mAFP is in place, individualized management comprises 3 essential components: maintenance, escalation, and deescalation. Figure 3 outlines a general approach to guide escalation and deescalation, while Table 3 summarizes key best practices.

Maintenance

Following mAFP placement, frequent monitoring of hemodynamic, laboratory, and clinical parameters is essential to ensure optimal support. The power level (P-level) should be titrated to balance effective decongestion and improved systemic perfusion against adverse effects that become more likely at higher settings (eg, hemolysis, suction events, and acute kidney injury).⁵⁷ Inotropes and vasopressors may be used concurrently to achieve hemodynamic targets. Bedside indicators of an appropriate P-level include warm extremities, capillary refill <2 seconds, improving congestion (eg, pulmonary/peripheral edema), improved mental status, robust urine output, and progressive reduction in vasopressor requirements. Serial laboratory assessment should reflect improving markers of perfusion and congestion, including central venous oxygen saturation, serum creatinine, aspartate aminotransferase/alanine aminotransferase, bilirubin, and lactate, and should include daily screening for bleeding/hemolysis with a complete blood count, lactate dehydrogenase, plasma free hemoglobin, and haptoglobin (Table 2). Although patients are supported with mAFPs, PACs are particularly useful for invasive optimization, guiding reductions in filling pressures (eg, lower pulmonary capillary wedge pressure), evaluating for evolution of shock phenotype, including subsequent vasodilatory shock, and ensuring adequate cardiac output measured by thermodilution or the Fick method.

Escalation

Failure to meet hemodynamic and perfusion targets should prompt first-line optimization by increasing the

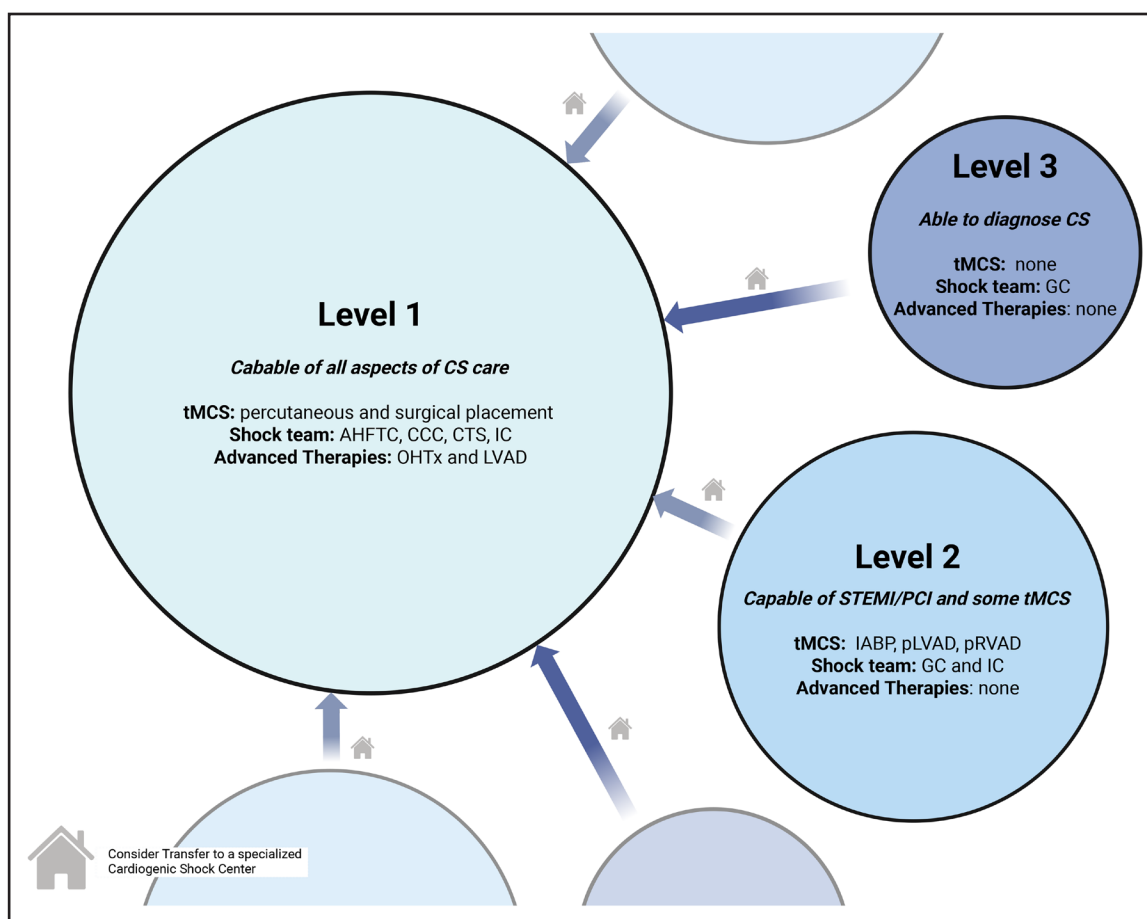


Figure 2. Proposed classification of cardiogenic shock (CS) centers based on locally available resources (proposed by Sinha et al 2025).⁵¹

Patients with CS may present at centers with varying infrastructure and resources. Transfer to a destination (Level 1) shock center should be considered by referral centers (Levels 2 and 3) after initial stabilization if more advanced tMCS is needed or candidacy for advanced therapies, including orthotopic heart transplant (OHTx) or left ventricular assist device (LVAD) need to be evaluated. AHFTC indicates Advanced Heart Failure and Transplant Cardiology; CCC, Critical Care Cardiology; GC, General Cardiology; IABP, intraaortic balloon pump; IC, Interventional Cardiology; LVAD, left ventricular assist device; RVAD, right ventricular assist device; and tMCS, temporary mechanical circulatory support. Created in BioRender. May, A. (2026) <https://BioRender.com/b7svwku>.

P-level and augmenting medical therapy. If an initial mAFP (ie, Impella CP) reaches maximal output without adequate hemodynamic, laboratory, or clinical improvement, upgrading to an axillary mAFP (ie, Impella 5.5) may be appropriate. Alternatively, persistent under-support despite mAFP placement, increasing oxygenation requirements, progression to biventricular failure, or development of complications (such as limb ischemia, bleeding, or hemolysis) should prompt escalation to alternative configurations (eg, veno-arterial extracorporeal membrane oxygenation or ECPELLA).²³

Deescalation

Once there is objective evidence of myocardial recovery, there should be consideration for weaning mAFP support. Timing and weaning rate should be individualized to cause and clinical trajectory. For example, patients with AMI-CS after successful early

revascularization may tolerate faster weaning than those with acute-on-chronic HF-CS, who often require a more gradual approach. Although weaning protocols vary, close monitoring of hemodynamic, laboratory, and clinical parameters is universally essential.^{58,59} At our institutions, we often wean mAFP support (ie, lowering the P-level) while maintaining low-dose vasopressor or inotropic support to facilitate earlier device removal and reduce device-related complications. Invasive hemodynamic monitoring equipment (eg, PAC, arterial line) and other hemodynamic monitoring tools (eg, Impella's SmartAssist monitoring system [Figure 4]) may be used to help guide weaning. A significant rise in vasopressor or inotrope requirements during mAFP weaning indicates failed weaning and warrants a return to the previous level of support. Persistent hemodynamic deterioration during repeated deescalation attempts should prompt reassessment of candidacy for advanced HF therapies.

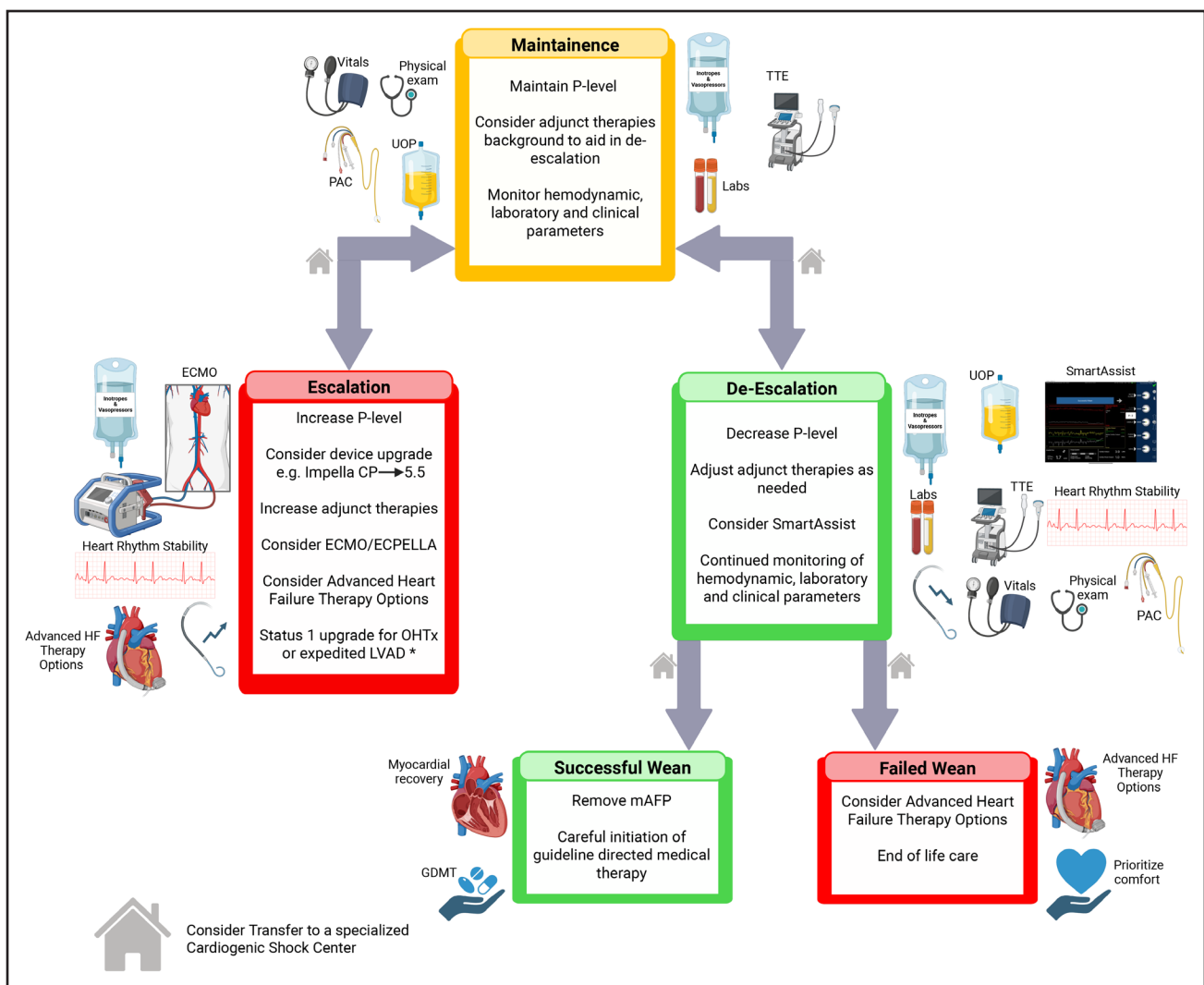


Figure 3. Proposed maintenance, escalation and deescalation strategy for patients with Impella therapy for management of cardiogenic shock (CS).

After Impella placement, diligent hemodynamic, laboratory, and clinical monitoring is essential to guide appropriate P-level selection and the use of adjunct therapies (eg, inotropes and vasopressors) to maintain optimal hemodynamics. If monitoring indicates poor perfusion or worsening congestion, escalation may be required. Options include upgrading to a higher-powered microaxial flow pump (mAFP; eg, Impella 5.5), transitioning to extracorporeal membrane oxygenation (ECMO)/ECMO+LV Impella (ECPELLA), or expedited left ventricular assist device (LVAD) consideration in eligible patients. Conversely, if objective signs of myocardial recovery emerge on serial monitoring, stepwise P-level reduction with continued close surveillance is appropriate. Following a successful wean, mAFP removal should be accompanied by careful initiation of guideline-directed medical therapy (GDMT). If weaning fails, advanced therapies such as orthotopic heart transplant (OHTx) or LVAD should be pursued in eligible patients, while palliative and end-of-life care should be prioritized for those not suitable for advanced interventions. PAC indicates pulmonary artery catheter; TTE, transthoracic echocardiography; and UOP, urine output. Created in BioRender. May, A. (2026) <https://BioRender.com/e2m3q5g>.

Inability to Wean Microaxial Flow Pump Support

Failure to wean from mAFP support is common and should prompt evaluation for durable LV assist device implantation or orthotopic heart transplantation. Such assessments, and any subsequent interventions, should be performed at a level 1 or advanced HF center (Figure 2). For patients not already at such a center, early transfer to a facility capable of providing advanced HF therapies and possessing expertise in hemodynamic

recovery and tMCS weaning is preferred. In these settings, patients can undergo timely advanced-therapy evaluation; receive expert multidisciplinary care to optimize mechanical support and adjunctive therapies (eg, complex PCI, structural interventions, electrophysiology procedures); and, when feasible, achieve separation from temporary support. For patients who are ineligible for advanced therapies and cannot be separated from mAFP support, goals-of-care discussions should be undertaken, and palliative care options should be offered.

Table 3. Best Practices for mAFP Management: Anticoagulation, Dressing Integrity, Neurovascular Assessment, Placement Verification, Pressure Waveform Analysis, SmartAssist Utilization, and Special Considerations During Cardiac Arrest

Best practices for microaxial flow pump management	
Systemic anticoagulation	Unfractionated Heparin (most common) Bivalirudin or Agatroban (in cases of HIT or elevated bleeding risk) aPTT goal 55–80 s or anti-Xa range 0.2–0.4 IU/mL* may vary by institution, more conservative targets may be considered for bleeding patients
Dressing assessment	Daily evaluation of dressing integrity Daily evaluation of bleeding, oozing, and erythema
Neurovascular assessment	Frequent neurovascular assessments to evaluate perfusion, sensation, and motor function in the extremity distal to the mAFP Deterioration of neurovascular status should prompt closer monitoring and consideration of an alternative tMCS strategy
Placement assessment	Bedside echocardiography Perform echocardiography to assess placement Impella CP: Inlet in the LV 3.5 cm from the aortic valve Impella 5.5: Inlet in the LV, 5 cm from the aortic valve Waveform Analysis LV waveform is estimated should resemble a ventricular waveform Aortic waveform is sensed by optical sensor and should resemble an aortic waveform Possible malposition ought to be evaluated by bedside echocardiography and may be reflected by: Overlap of LV and Aortic waveforms Lack of motor current pulsatility
Pressure waveform analysis	Negative LV diastolic pressure may indicate increased risk of suction alarms due to: Hypovolemia RV failure Aortic pressures displayed on Impella console should approximate what is measured (eg, blood pressure measurements via arterial line)
Data trend functionality (eg, Smart Assist)	Input CO (via Fick or thermodilution) data into the Impella console at least every 8 h Observe trends in the following parameters and use them to guide decision-making regarding escalation or deescalation: Aortic pressure LVEDP CPO CO components Overall CO Impella contribution to CO Native ventricle contribution to CO
Cardiac arrest	Cardiac arrest: Perform CPR per standard ACLS protocol OK to maintain P-level (at prearrest levels) during CPR If ROSC is achieved: Perform echocardiography to ensure appropriate Impella placement following chest compressions Escalate support as needed and as tolerated, consider need for escalation to VA-ECMO/ECPELLA

ACLS indicates advanced care life support; aPTT, activated partial thromboplastin time; CO, cardiac output; CPO, cardiac power output; CPR, cardiopulmonary resuscitation; ECPELLA, VA-ECMO+LV Impella; HIT, heparin-induced thrombocytopenia; LV, left ventricular; LVEDP, left ventricular end-diastolic pressure; mAFP, microaxial flow pump; ROSC, return of spontaneous circulation; RV, right ventricular; tMCS, temporary mechanical circulatory support; and VA-ECMO, veno-arterial extracorporeal membrane oxygenation.

*aPTT and anti-Xa goal ranges may vary by institution.

COMPLICATIONS OF MICROAXIAL FLOW PUMPS: SURVEILLANCE, PREVENTION, AND MANAGEMENT

Even as the benefits of mAFP are realized in appropriately selected patients, vigilant surveillance for and mitigation of complications (Table 4) remain essential to achieving optimal outcomes.

Available evidence indicates that mAFP use carries a substantial adverse-event burden. In 1 retrospective study, 93% of patients receiving left-sided Impella support experienced at least 1 complication.⁶⁰ In the multicenter DanGer Shock trial, the composite safety end point—including severe bleeding, hemolysis, limb ischemia, device failure, or worsening aortic regurgitation—occurred in 24.0% of patients. Moreover, the risk of

acute kidney injury requiring renal replacement therapy was also higher with mAFP compared with standard care alone (49.1% versus 26.7%).⁵ Given the appreciable risk of device-related complications, CS management with mAFP should include proactive strategies tailored to each patient's risk profile and clinical trajectory.

Bleeding

Bleeding is a major complication of mAFP support. Risk is heightened by large-bore arterial access, the need for systemic anticoagulation, and acquired von Willebrand factor deficiency.^{76,77} Heparin is the most commonly used systemic anticoagulant but increases the risk of bleeding and heparin-induced thrombocytopenia. In patients at higher risk of bleeding or heparin-induced

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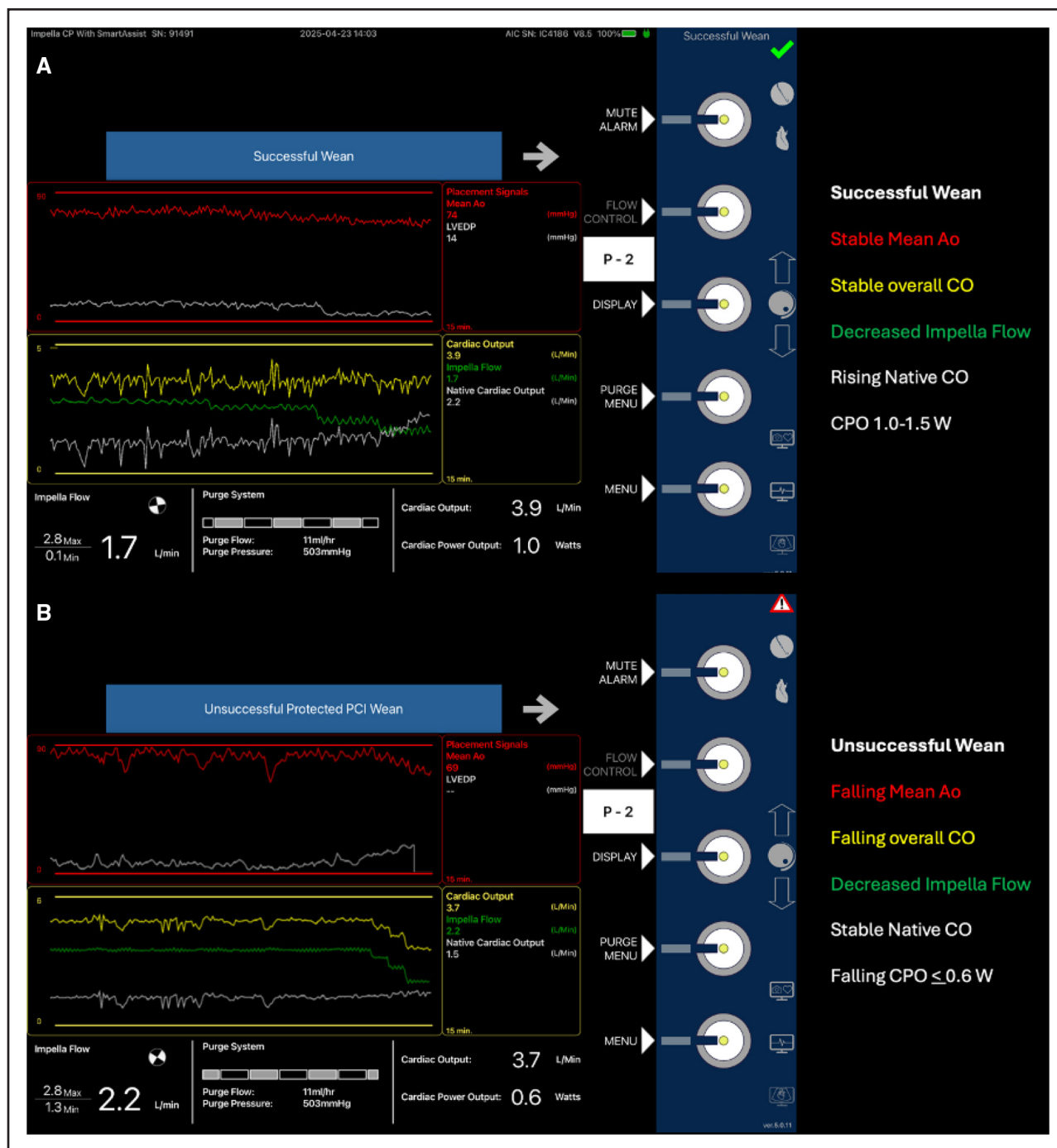


Figure 4. The SmartAssist Trend Screen supports personalized decision-making during Impella weaning.

We find the Impella SmartAssist dashboard particularly useful for adjudicating weaning success. The dashboard displays estimated mean aortic pressure, estimated cardiac power output (CPO), estimated left ventricular end-diastolic pressure (LVEDP), and estimated pump flow (Impella flow). When total cardiac output (CO)—measured by thermodilution or the Fick method at baseline and after each P-level adjustment—is entered into the system, the console automatically calculates native CO (total CO–Impella flow) and tracks trends over time. During stepwise reductions in P-level, 3 parameters should be followed: total CO, native CO, and Impella flow. A successful wean is characterized by stable mean aortic pressure and total CO, accompanied by a rise in native CO as Impella flow decreases (A). In contrast, an unsuccessful wean is indicated by a decline in mean aortic pressure and total CO without a compensatory increase in native CO as Impella support is withdrawn (B). In addition, as the Impella flow decreases, an increased arterial systolic and pulse pressure is expected, and a low pulse pressure during Impella weaning may herald failure to wean. Ao indicates aortic pressure. Created in BioRender. May, A. (2026) <https://BioRender.com/hedznp9>.

thrombocytopenia, bicarbonate-based purge solutions and systemic anticoagulation with short-acting direct thrombin inhibitors (eg, bivalirudin) may be feasible.^{78,79}

A range of biological, physiological, and therapeutic factors contribute to bleeding risk during tMCS placement and subsequent intensive care unit care, and risk mitigation requires adherence to safe vascular practices, including fluoroscopy, ultrasound-guided access, femoral angiography, securement of large-bore sheaths, and confirmation of distal vessel perfusion.⁸⁰ Postoperative care must prioritize serial assessment of anticoagulation status, access-site bleeding, and distal limb perfusion, and actively balance pharmacotherapy to weigh the risk of bleeding versus thrombosis.⁸⁰ In a decompensating patient, postoperative complications (eg, retroperitoneal hematoma and pseudoaneurysm) must also be considered.

Hemolysis

Intravascular hemolysis is a significant complication of mAFPs that can lead to pigment-induced acute kidney injury, anemia, and tissue ischemia.⁶⁸ Its incidence increases with higher pump speeds (P-levels), device malposition, and pump thrombosis.^{64,81,82} Because device thrombosis both triggers and amplifies hemolysis, maintaining systemic anticoagulation and using appropriate purge solutions are essential.⁶⁵ Hemolysis should be suspected in patients with worsening anemia, elevated bilirubin, and hemoglobinuria and can be confirmed by monitoring lactate dehydrogenase, haptoglobin, and plasma free hemoglobin. Plasma free hemoglobin may be more sensitive and specific than lactate dehydrogenase for identification of hemolysis in patients with transvalvular mAFP.⁸³ Once hemolysis is identified, it is important to confirm proper mAFP positioning within the ventricle using bedside echocardiography and make bedside adjustments as needed. For example, if the mAFP is positioned too deeply, with the device outlet at the level of the aortic valve, turbulent flow may occur as blood is expelled from the device at this site, increasing shear stress and promoting hemolysis. It is also important to acknowledge that hemolysis risk scales with mAFP motor speed and is therefore more common with smaller mAFPs (eg, Impella CP) than with larger-bore models (eg, Impella 5.5).⁶⁵ In addition to device-related factors that may increase hemolysis, several patient-centered characteristics have also been associated with a higher likelihood of laboratory-defined hemolysis, including emergency device insertion, lower baseline ejection fraction, AMI as the cause of CS, higher baseline lactate, higher heart rate, and lower pulse pressure.⁸⁴ Mitigation measures include reducing the P-level to the lowest effective setting and, when necessary, upsizing to a larger Impella device with lower rotational speeds.

Device Thrombosis

Device thrombosis represents a serious complication that predisposes patients to deleterious downstream events, including hemolysis and embolic stroke. This risk is generally mitigated through appropriate systemic anticoagulation. Practical indicators of device thrombosis include warnings of rising purge pressure with decreased purge flow on the mAFP monitoring system. Additional red flags include laboratory or clinical evidence of hemolysis or embolic phenomena (eg, stroke).

When thrombosis is suspected or confirmed, management may require device removal and transition to alternative tMCS.⁶⁴ Small case reports describe a salvage strategy of adding tissue plasminogen activator to the purge solution to lyse thrombus without major bleeding.^{75,85} We have used this approach successfully at our institutions; however, its safety and efficacy require further investigation before it can be recommended routinely.

Critical Limb Ischemia

Critical limb ischemia is a serious complication that can result from arterial thrombus formation on the catheter, distal embolization, or direct arterial occlusion from the large-bore sheath/device. Preexisting peripheral artery disease is an independent predictor of acute limb ischemia in this setting.⁶⁹ Risk mitigation hinges on frequent neurovascular assessments and minimizing interruptions in anticoagulation. Placement of a distal perfusion catheter at the time of Impella insertion may be a potential option to further mitigate risk.⁷¹ If critical limb ischemia is identified, prompt involvement of peripheral vascular specialists (surgical or interventional) is necessary, as removal of the offending device or other vascular interventions may be required to restore limb perfusion. When ongoing hemodynamic support is needed, device removal should be coordinated with immediate establishment of alternative access for tMCS; for example, conversion from a transfemoral Impella CP to an axillary Impella 5.5.

Stroke

Stroke is a less frequent, but serious complication associated with mAFP use.⁸⁶ Exposure to anticoagulation or the development of heparin-induced thrombocytopenia increases the risk of intracranial hemorrhage.⁸⁷ Conversely, interruption of systemic anticoagulation or the development of device thrombosis can lead to cardioembolic stroke. Patients at higher risk for stroke include those requiring prolonged support, thrombocytopenia, inherent coagulopathy, recent neurological surgery or trauma, or those needing anticoagulation interruption. For patients with an elevated risk of hemorrhagic stroke, it may be beneficial to adopt more conservative activated partial thromboplastin time targets or to choose

Table 4. Complication Frequency, Monitoring, and Considerations in Mitigating and Addressing Each Complication

Complication	Rate of occurrence	Monitoring	Mitigate	Address
Hemolysis	7.5–41.6% ^{60–63}	↑ LDH ↓ Haptoglobin ↑ Plasma free Hemoglobin ↑ Bilirubin ↑ Urobilinogen Urine discoloration (pink or merlot-colored)	Less hemolysis with larger Impella 5.5 due to lower rotational speed in comparison to smaller Impella CP ^{64,65} Consider adjusting depth/position of Impella	Lower rotational speed (ie, reduce P-level) Check and optimize device position Consider device thrombosis Consider device removal vs escalation to alternative device ⁶⁴
Major bleeding	0.05–62.3% ^{9,60–63,66}	↓ Hemoglobin Access-site evaluation (eg, obvious bleeding at access site) Hemodynamic monitoring (eg, overlapping hemorrhagic shock profile)	Optimizing femoral cannulation technique: ultrasound guidance, micro puncture needle use, vessel closure device, adequate training in large-bore closure techniques ⁶⁷ Consider patient-specific risk factors that increase risk for bleeding	Consider using anticoagulant with shorter half-life (bivalirudin) Consider transition to BBPS only If anticoagulation must be interrupted to achieve hemostatic control, increase pump speed to maximally tolerated ⁶⁸
Critical limb ischemia	0.1–12.8% ^{60,62,63,66}	Abnormal neurovascular assessment (eg, cool and pale leg) ↑ Lactate ↑ AST ↑ CK ↑ LDH ↑ Potassium ⁷²	If significant PAD is present, consider use of IABP when able or axillary placement of Impella ⁶⁹ Ensure therapeutic systemic anticoagulation and avoid interruptions in anticoagulation	Involve vascular surgery urgently for surgical hemostasis and removal of Impella ⁷⁰ Optimize and increase CO and MAP Explore alternative vascular access for implantation of new tMCS Consider placement of distal perfusing catheter ⁷¹
Stroke	1.7–6.5% ^{60,61,63,66}	Neurological assessment (eg, new neurological deficits)	Avoid anticoagulation interruption Consider patient-specific risk factors that increase risk for bleeding Consider use of lower aPTT goals in high-risk patients	Urgent involvement of neurology ± neurosurgery Use of systemic tPA for mAFP patients with ischemic stroke is not well established
New acute valvulopathy	3.2–6.4% ^{60,73}	Physical exam (eg, new murmur) Echocardiography (eg, flail mitral valve) Changes invasive hemodynamic monitoring parameters (eg, drop in MAP and diastolic blood pressure for acute aortic regurgitation)	If sudden deterioration in hemodynamic stability, echocardiography is warranted	Avoid high afterload states Consider valve repair/replacement
Infection	1.8–14.3% ^{63,66,74}	↑↓ Temperature ↑ Mixed venous oxygen saturation ↑ Leukocytes (+) Blood cultures ↓ SVR ↓ RAP Access-site evaluation (eg, grossly visible signs of CLABSI [eg, erythema])	Perioperative antibiotics Access-site care Remove unnecessary indwelling lines as soon as able	Involve Infectious Disease team Targeted antibiotics If debridement is needed involve relevant surgical teams Remove unnecessary indwelling lines Replace necessary indwelling lines Device removal if severe sepsis, septic emboli, suppurative thrombophlebitis, or persistent bacteremia >3d ⁶⁴
Device thrombosis	4.4–15.2% ^{63,64}	Echocardiography (eg, visible catheter-associated thrombus) ↑ Purge pressure ↓ Purge flow Suspect if evidence of: Hemolysis Stroke Limb ischemia	Consider increased aPTT/ACT target or addition of direct thrombin inhibitor ⁶⁴	Incorporation of tPA into purge fluid as a salvage technique has been seen in few case reports, however more research is needed to determine efficacy and ensure patient safety ^{64,75} Device removal and replacement if needed with tMCS or durable LVAD ⁶⁴

ACT indicates activated clotting time; aPTT, activated partial thromboplastin time; AST, aspartate aminotransferase; BBPS, bicarbonate-based purge solution; CK, creatine kinase; CLABSI, central line-associated bloodstream infection; CO, cardiac output; IABP, intraaortic balloon pump; LDH, lactate dehydrogenase; LVAD, left ventricular assist device; mAFP, microaxial flow pump; MAP, mean arterial pressure; PAD, peripheral arterial disease; RAP, right atrial pressure; SVR, systemic vascular resistance; tMCS, temporary mechanical circulatory support; and tPA, tissue plasminogen activator.

anticoagulants with shorter half-lives (eg, bivalirudin in patients without renal failure).

Infection

Due to the invasive nature of mAFP, localized access-site infections and bloodstream infections are significant concerns for all patients. Risk factors include a high body mass index, diabetes, chronic kidney disease,

and renal replacement therapy.⁸⁸ Differentiating device-related bloodstream infections due to the device or from other sources of infection, such as pneumonia, urinary tract infections, or central line-associated bloodstream infections, is essential for appropriate management. Antibiotic selection should be guided by the most likely pathogens within the clinical context and aligned with institution-specific antibiograms; for device-related bloodstream infections, consideration of the potential

presence of biofilm formation is important in antibiotic choice.^{64,89} Device removal is recommended in cases of severe sepsis, persistent bacteremia, or septic emboli.⁶⁴

Acute Valvulopathy

Valvular complications with mAFPs are uncommon, but can involve the aortic and mitral valves.⁹⁰ Transient, low-grade aortic regurgitation may occur from leaflet mal-coaptation and increased afterload during support and typically resolves after device removal. New, persistent aortic regurgitation after mAFP catheter placement is rare (eg, 3.2% in 1 study) and may result from valve trauma during insertion, repositioning, or removal.^{73,91} Mitral valve complications generally reflect chordal injury during placement, leading to mitral regurgitation.^{73,90} Any emergent or unexplained hemodynamic deterioration despite mAFP support should prompt urgent echocardiographic evaluation for valve-related complications.

CLINICAL RESEARCH PRIORITIES IN MAFP PERSONALIZATION

While this review highlights important considerations in mAFP selection and management, it is clear that several knowledge gaps remain regarding mAFP personalization across various phenotypes. Table 5 outlines potential research directions across multiple clinical domains.

Lessons Learned Leading to Phenotype-Stratified Clinical Trials

The discordant results of randomized controlled trials of tMCS in CS have pointed toward a need for more precise phenotypic stratification.^{92–94} Broad inclusion criteria, enrollment of patients with nonmodifiable conditions (eg, anoxic brain injury), a narrow focus on AMI-CS, and small sample sizes are potential explanations for the

absence of treatment effects across multiple trials. In response, more recent trials have attempted to address these limitations.

The DanGer Shock trial (2024) used tighter inclusion and exclusion criteria to create a more homogeneous AMI-CS cohort and became the first randomized controlled trial to show a survival benefit of an mAFP (Impella CP) versus standard care among selected patients with STEMI-CS.⁵ However, due to stringent enrollment criteria, its cohort represented only about 5% of all CS presentations and 32% of patients with STEMI-related CS.⁹⁵ The trial deliberately excluded patients with preserved LV function, late presentation, out-of-hospital cardiac arrest, overt RV failure, non–ST-segment–elevation myocardial infarction, and HF-CS; thus, the observed benefit should not be extrapolated to these groups. Although rigorous enrollment criteria can refine a heterogeneous population into a well-defined subset and strengthen the ability to detect true treatment effects, this approach inevitably limits generalizability.

In addition, the recently published, but underpowered ALT-SHOCK-2 trial (Study on Early Intra-aortic Balloon Pump Placement in Acute Decompensated Heart Failure Complicated by Cardiogenic Shock; 2025) used a phenotype-stratified design in HF-CS but found no difference in 30-day mortality, overall survival, or successful bridging to heart replacement therapy between intraaortic balloon pump and standard care.⁹⁴ However, the trial operationalized routine intraaortic balloon pump use and included a substantial cohort of SCAI stage B patients who may have been unlikely to benefit from tMCS. PACs were used in <50% of subjects in either arm, leaving most participants without invasive hemodynamics to guide personalized titration of mechanical or pharmacological support. In addition, considerable crossover and the small sample size may have blunted any efficacy that might have emerged with more selective tMCS use. Collectively, these trials signal a deliberate shift toward more selective, phenotype-driven, and tailored deployment of tMCS.

Table 5. Research Priorities Across Various Clinical Research Domains

Clinical research domains	Priorities
RCTs	Define a common language for all CS research pertaining to tMCS (eg, SHARC) to improve cross-study comparability
	Conduct RCTs that have the greatest impact on meaningful patient subgroups
Clinical phenotyping	Conduct dedicated studies to discover and characterize phenotypes in patients with CS
	Identify how meaningful phenotypic patterns influence treatment options (eg, tMCS selection)
Advanced analytics	Use machine learning to integrate objective clinical parameters and identify clinical phenotypes
	Develop rapidly deployable clinician decision-support tools
	Create smart escalation and weaning algorithms through real-time interpretation of objective data
	Determine how artificial intelligence tools can be integrated into CS management and distributed across hospital systems
Patient safety/quality improvement	Prioritize research aimed at optimizing current tMCS performance, durability, and complication mitigation
	Identify patient phenotypes for whom the risk of tMCS outweighs the benefit

CS indicates cardiogenic shock; RCT, randomized control trial; SHARC, Shock Academic Research Consortium; and tMCS, temporary mechanical circulatory support.

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Priorities in Clinical Research

As future trials increasingly use selective enrollment strategies, additional dedicated prospective studies in well-defined populations will be crucial to establish whether a true benefit exists. Moreover, as phenotyping becomes more integrated into clinical research, efforts are needed to address major gaps in understanding the biological relevance of each phenotype and its implications for tMCS selection. Despite these limitations, precise identification of patients most likely to benefit is worthwhile given the substantial cost, procedural risk, and specialized care associated with tMCS. Going forward, the most effective CS care may hinge less on further technological advances in tMCS and other therapies, and more on deploying existing therapies in the right patients at the right time.

To this end, the adoption of standardized terminology (eg, SCAI staging) is a necessary first step to enhance prospective trial comparability and clinical applicability. In support of this, the Shock Academic Research Consortium consensus provides unified definitions for CS, patient populations, and outcomes—including safety end points—thereby establishing a common language and consistency for academic CS and tMCS research.⁹⁶ Through the use of shared definitions and terminology, we can more effectively evaluate the safety and efficacy of tMCS for a wide variety of CS phenotypes.

Identifying rapidly deployable, cost-effective tools that integrate objective parameters to evaluate patients with CS will be crucial for accurately determining a patient's CS phenotype. Targeted studies to delineate patient phenotypes may be necessary to characterize specific CS subgroups more precisely. Given the prevalence of negative and neutral RCTs, potentially driven by CS heterogeneity, it may be advantageous to design trials that prioritize subgroups with the greatest potential for meaningful patient impact.

Potential for Advanced Analytics in CS Management and tMCS Optimization

Modern CS care increasingly requires real-time interpretation of and adaptation to multidimensional, continuously evolving data. Given the heterogeneity and dynamic nature of CS, clinicians must rapidly acquire, categorize, and synthesize information from physical examination findings, laboratory data, echocardiography, and left- and right-heart catheterization. This iterative process blends subjective assessment with objective metrics to deliver personalized care.

Meeting this complexity will likely demand moving beyond broad, simplistic categorization frameworks. Iterative modeling and advanced analytics using machine learning (ML) and other artificial intelligence techniques offer an alternative path forward through phenotyping,

prognostication, and identification of treatable traits.⁹⁷ Although there is currently no evidence to suggest that ML improves CS outcomes, by assimilating high-dimensional data and organizing patient phenotypes, novel ML methods may help identify which patients are more or less likely to benefit from specific treatment strategies, including mAFP.

Notably, Zweck et al¹² used ML to identify 3 CS phenotype clusters based on hemodynamic and laboratory data: noncongested, cardiorenal, and cardiometabolic. Their findings suggest that the cardiometabolic profile had the highest mortality and most frequent presentation with advanced SCAI stages. In follow-up work, the same group reported that, in a nonrandomized observational analysis, tMCS use in the cardiorenal phenotype was associated with higher mortality compared with the cardiometabolic and noncongested phenotypes.¹³ These findings, among others, support the hypothesis that ML-based phenotyping may ultimately prove useful not only for risk stratification but also for identifying treatable traits that enable more precise and effective therapy selection.⁹⁸

Looking ahead, we anticipate that ML techniques and artificial intelligence technologies will become essential for real-time, personalized decision support in CS. It will be increasingly important that future analytic tools be designed and implemented with explicit consideration of timing, as initiating therapy either too early or too late may diminish potential benefit. Until these tools are fully developed, prospectively validated, and broadly deployable at the bedside, clinicians must continue to rely on expert judgment to interpret complex data and guide care.

Addressing Patient Safety and Quality Improvement

As discussed, mAFP utilization is associated with high complication rates. Quality improvement and patient-safety research focused on complication monitoring, prevention, and mitigation are essential not only for patient safety but also for improving the durability of mAFPs. The ongoing OASIS-AMICS trial (Observational Assessment of Support with Impella Best Practices in Acute Myocardial Infarction Complicated by Cardiogenic Shock; REGISTRATION: URL: <https://clinicaltrials.gov/study/>; Unique identifier: NCT06964685) will evaluate safety outcomes in patients with AMI-CS supported with Impella CP during PCI and may help identify effective strategies to mitigate complications associated with mAFPs. Novel quality-improvement initiatives addressing anticoagulation management in the contexts of bleeding, thrombosis, and critical limb ischemia are prudent, as these scenarios represent some of the highest complication rates. In addition, standardized approaches to mitigating complications (eg, anticoagulation strategy)⁹⁹ may

reduce variability in complication rates across healthcare systems. Although efforts to address this in the high-risk PCI population have been proposed,⁶⁷ further research is needed to address this issue in the CS population, where mAFP use may be required for prolonged periods.

CONCLUSIONS

CS is a complex, heterogeneous syndrome that requires individualized management. As demonstrated in the DanGer Shock trial, when tailored to appropriately selected patients with STEMI-related CS, mAFP has improved outcomes. Although evidence is limited, mAFP may also prove beneficial in other CS populations for whom high-quality data are not yet available. Accordingly, mAFP use warrants careful appraisal of shock severity and cause, hemodynamic profile, and clinical trajectory, coupled with a clearly defined therapeutic intent and contingency plan should tMCS prove ineffective or weaning fail. Successful mAFP use depends not only on appropriate initiation but also on meticulous management, timely escalation or deescalation, and vigilant surveillance with proactive mitigation of complications. These efforts are best coordinated by a dedicated multidisciplinary shock team within specialized centers operating as part of a regionalized system of care.

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